

Sachin Ramanathan

858-214-0969 | saramanathan@ucsd.edu | github.com/sachin-dot-py

EDUCATION

University of California San Diego

Bachelor of Science in Computer Science (GPA: 4.0)

San Diego, CA

Sep 2023 – Dec 2026

WORK EXPERIENCE

Website Designer Programmer

Jan 2024 – Present

Qualcomm Institute, UC San Diego

San Diego, CA

- Revamped a web application using React.js & TypeScript, deploying weekly updates using a CI/CD pipeline.
- Developed and containerized a Next.js web app using Docker and Kubernetes, enabling seamless scalability.
- Collaborated with educators to tailor the web app for K-12 classrooms, used by 500+ students and teachers.
- Created a Python-Flask REST API, interfacing with a supercomputer to run complex molecular simulations.
- Automated recurring website maintenance tasks using Python scripts, improving operational efficiency by 200%.

Student Affairs Assistant

Jan 2024 – Present

Halicioglu Data Science Institute, UC San Diego

San Diego, CA

- Facilitated large-scale industry events with 100+ participants, assisting in event setup and logistics.
- Managed and triaged 200+ emails weekly across shared inboxes, improving response efficiency by 50%.

Intern Software Developer

Nov 2021 – Mar 2022

Jagan's Group of Institutions

Remote

- Engineered a full-stack student information system for 2000+ students, reducing record modification time by 80%.
- Developed the web application using Python (Flask, Jinja2) for the backend, an SQL database for data storage, and HTML, CSS, and JavaScript for the front end, with Git for version control and GitHub for collaboration.
- Designed prototypes and conducted iterative user testing, integrating feedback to enhance user satisfaction by 30%.

PROJECTS

IGCSE Discord Bot | TypeScript, MongoDB, Redis, Python

[View on GitHub](#)

- Led a team of 15 devs to develop a Discord bot used by 66,000+ users across 450 communities with 99% uptime.
- Enabled students to study better with question banks and quizzes on exam topics, with fast response times (<1s).

Illuminate | React.js, Python, Flask

[View on GitHub](#)

- Developed a web app for UCSD students to automate planning classes, reducing course planning time by 80%.
- Designed efficient algorithms to select the top three optimal schedule combinations from thousands of possibilities.
- Built a web-scraping agent using Python to parse and extract real-time course data from UCSD's website.

FoodShare | Python, Flask, MySQL, HTML, CSS, Javascript

[View on GitHub](#)

- Developed a full-stack web app managing users, orders, and restaurant data, with OOP and a relational database.
- Integrated Google Maps API for location-based services, optimizing the display of restaurants by distance.
- Implemented an email notification system using smtplib, and utilised caching to reduce page load times by 70%.

LEADERSHIP EXPERIENCE

Lead Peer Mentor

Sep 2025 – Present

UC San Diego Computer Science and Engineering Department

San Diego, CA

AWARDS

4x Hackathon Winner: CalHacks (Berkeley), DiamondHacks (UCSD), Hackarama (LPL Financial), eYSRC (IIT Bombay)
Awarded the UCSD CSE Leadership Scholarship. Ranked Second Place at the UCSD Competitive Programming Contest.

TECHNICAL SKILLS

Programming Languages: Python, Java, React.js, Next.js, C, SQL, JavaScript, TypeScript, HTML, CSS, Bash.

Databases/Frameworks: MongoDB, MySQL, Flask, Pandas, Tkinter, Matplotlib, Selenium Automation.

Developer Tools: Git, GitHub, Docker, AWS, Bash, Cron, Vim, VS Code, Linux.

Certifications: Completed certification course on Agile Software Development (Scrum, Kanban, XP).

Concepts: Data Structures and Algorithms, Object Oriented Programming, REST APIs, CI/CD, Databases, GenAI.